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Studierichting: Informatica
Oefeningenreeks 4A nr. 14.2

$$\int x \ln x dx$$

$$\left(\int udv = uv - \int vdu\right)$$

$$\begin{cases} u = \ln x \\ dv = xdx \end{cases} \Rightarrow \begin{cases} du = \dfrac{1}{x}dx \\ v = \dfrac{1}{2}x^2 \end{cases}$$

$$= \ln x \cdot \frac{1}{2}x^2 - \int \frac{1}{2}x^2 \cdot \frac{1}{x}dx$$

$$= \frac{1}{2}x^2 \ln x - \frac{1}{2}\int xdx$$

$$= \frac{1}{2}x^2 \ln x - \frac{1}{4}x^2 + C$$

$$= \frac{x^2 \ln x}{2} - \frac{x^2}{4} + C$$

References